SmartPayEnergy: Intelligent IoT Prepaid Electricity Monitoring system

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***Abstract*—The rapid development of smart home technologies has made the necessity of efficient and real-time energy monitoring systems even more important. The paper is a Smart Energy Monitoring system of Wi-Fi based that is developed to estimate and show electrical parameters used in residential purposes. The system has been built on ESP32 microcontroller with a PZEM-004T energy sensor which is used to measure voltage, current, power, and energy use with high precision.**

The collected data is sent straight away via the in-built Wi-Fi support of the ESP32 to a web-based dashboard interface in order to visualize it on a real-time basis. Dashboard shows real-time electrical values and past trends of energy use to allow users to track their energy consumption.

Besides monitoring, the system has a balance estimation feature, in which the energy cost is computed depending on the amount of total energy used and shown on the dashboard. This will enable the users to monitor their estimated balance left and be more careful with the use of electricity. The suggested system is economical, can be scaled and ideally applicable in smart residential energy awareness and energy monitoring.

***Index Terms*—Index Terms- Energy monitoring, ESP32, PZEM-004T, Smart home, Wi-Fi communication, Dashboard, Power consumption.**

I. INTRODUCTION

The high-paced development of the smart home technologies and rising electricity demand have resulted in the need to have effective and real-time energy monitoring systems. To enhance energy awareness, minimize wastage and safe functionality of residential electrical system, performance of electrical parameters including voltage, current, power and consumption of electricity is important. The old type of electricity meters are cumulative only and do not have real time monitor or easy visualization meaning consumers cannot effectively monitor and manage their energy consumption.

The Internet of Things (IoT) has brought about a method of monitoring systems that are based on microcontrollers, thus making them an affordable and scalable approach to smart energy. The ESP32 microcontroller is one of these and has become very popular as it has an inbuilt Wi-Fi, low power consumption and compatibility with sensors. The energy measurement modules like PZEM-004T allow proper measurement of electrical parameters like the voltage, current, power, and total energy consumed.

In this paper, a Smart Energy Monitoring System based on a Wi-Fi system is created with an ESP32 microcontroller and a PZEM-004T energy sensor. The measured data is sent via Wi-Fi and shown on a web-based dashboard interface which enables users to view the electrical parameters in real time. The system allows constant monitoring of energy use and makes the user comprehend their energy consumption patterns.

There is a balance estimation feature besides monitoring in real time. The approximate cost is obtained and shown on the dashboard based on the sum of energy used and a predetermined unit cost of electricity. This attribute improves the awareness of users, making it easy to get a clear picture of the amount of electricity that is spent and thus the need to promote energy saving behaviour.

The system suggested is cheap, simple to install, and applicable in residential smart energy monitoring applications. It offers a viable move towards smart energy management through a combination of precise sensing, wireless communication, and visualization that is easy to use.

II. LITERATURE SURVEY

The most recent research on the AI-based energy consumption forecasting indicates that when the IoT data is combined with the advanced machine learning models, the forecasting accuracy is higher than with the conventional forecasting models. The older methods such as ARIMA, and linear regression are typically only 70-80 percent accurate, and MAPE is often greater than 15 percent due to the fact that they do not capture nonlinear trends due to changes in weather, occupancy and load. The modern algorithms (Random Forest, SVR, Gradient Boosting, and deep neural networks (LSTM, GRU, CNN)

achieve a 20-35% error reduction and frequently provide MAPE with values of 5-8. Good preprocessing of data involves missing values, outlier removal and p feature selection only increases accuracy by 10-18%. IoT sensors working at high frequency that allow real-time data every 1-5 seconds allow more responsive and efficient energy-management systems [1].

As the paper suggests, a supervised machine learning algorithm to predict appliance-level energy consumption based on low-frequency smart-meter data demonstrates that accurate results can be obtained even with a sampling frequency of 110 Hz. This is achieved with the K-means clustering which classifies similar power values together and identifies the appliance operating state with 8592 percent state-classification accuracy. The trained model with the help of such features as active power (P), reactive power (Q) current (I), power factor (PF) achieves the overall prediction accuracy of 88-94 percent with 40-60 percent lower computational complexity than deep learning-based NILM methods. This renders the method very feasible to actual houses that have low hardware capacity. The authors also mention that the addition of dynamic or incremental learning in the future would help to enhance the adaptability and increase the performance by another 15-20, particularly when the patterns of household use change [2].

Appliance Load Monitoring is used to keep track of power consumption of appliances at home, industries and other systems. Conventional surveillance involves a large number of sensors and therefore is very expensive to install. Non-Intrusive Load Monitoring (NILM) proposes a less expensive solution, based on the total electricity measurement of a single main meter, and then attempts to determine the usage of individual appliances. Nevertheless, the performance of NILM is reduced in cases where appliances have a plurality of operating modes. An appliance-based NILM solution is suggested in this work to deal with the devices of various programs, and the various deep learning techniques are employed to increase the accuracy. These developed models were experimented in a laboratory based on an IoT microgrid setting [3].

According to the research, the combination of IoT and deep learning make the management of energy produced by smart grids much more robust as it facilitates real-time monitoring, predictive analytics, and automatics. As the demand of electricity in the world has increased more than 25 percent over the last decade, the conventional grids are not able to sustain stability and predictive load balancing is therefore needed to create less downtime and inefficiency. IoT devices produce gigantic real-time data, enhancing grid transparency and complementing forecasting models at up to 95% accuracy on predicting power demand, nearly 10 times better than the classical regression methods. The load balancing based on deep learning can also save energy by a factor of 20, increase grid stability by 15% and especially at times of peak consumption rate [4].

The consumption of electricity continues to increase in the world at an average rate of more than 3 per cent annually and residential and commercial areas, together abide almost 65 per cent of total demand and this imposes pressure on power infrastructure. As power prices vary by 20-40% and the world generation gap is estimated at 15%, precise short-term load forecasting (Grid Stability) short-term load forecasting (STLF) has become the key to grid stability. Experiments with a 168-hour forecasting horizon demonstrate that weather variation (which can double or triple the load) and holiday impacts are important factors to include in prediction to increase prediction accuracy. The contemporary ML techniques, especially XGBoost, results in the forecasting error of 3 to 5 percent, which is 5 to 10 percent lower than the traditional statistical estimators [5].

The paper proposes the use of IoT to enhance the efficiency and reliability of Smart Grids by allowing real-time monitoring of the demand and supply. The IoT sensors assist in demand forecasting, load balancing, and energy storage management to help minimise energy waste at the peak times. The system also allows a constant contact between users and suppliers, which makes delivery of power more cost effective and stable. It was also observed in the study that smart grids are based on IoT, which is associated with quicker response, the combination of distributed generation, and the ability to optimize the use of stored energy to a greater extent [6].

The need to fight for the increasing energy prices and the global energy demand necessitates smarter methods of monitoring and conserving power, particularly in homes. This new smart Energy Management System (EMS) addresses this through the Internet of Things (IoT) and Big Data analytics to overcome the constraints of the older home systems. Its methodology to transform the ordinary devices, such as the air conditioning units (which is a heavy energy consumer, up to 60 percent in areas such as the Arab Gulf), into a smart object by connecting small, high-speed microcontrollers (SoCs). These are mini monitors that continuously monitor the ambient conditions and power usage [7].

The project has revealed a proposed Internet of things (IoT) system of real-time monitoring and prediction of motor under different loading conditions of power consumption. With the further development of the IoT technology, the possibility that most of the systems will become more cultured and accommodating is high, which would require even more efficiency gains and cost reduction. By monitoring and analyzing continuously, implementation of energy saving measures can be determined and thus minimized costs and energy wastage in its best use. The timely reaction to any possible problems is also possible due to the real-time track of the power consumption, which will result in the enhancement of efficiency and productivity. On the whole, this project has added to the literature on the use of IoT to manage power and preconditions the development of further research in this field [8].

The paper is a review of energy management techniques of a sustainable smart city based on the Internet of Energy (IoE). It categorizes strategies such as the scheduling optimization, low-power device transceivers, cognitive frameworks and cloud computing integration. The paper identifies the importance of smart grids, microgrids, and nano grids in the attainment of efficient distribution of energy and discusses the

energy harvesting methods, optimal path of energy routing, and integration of multi-source. Other enabling technologies mentioned by the authors and a conceptual framework of the IoE-based energy management are suggested. Among major issues are the scaling, interoperability, and optimization of distributed resources, and the next step of technology in smart cities is to achieve better energy efficiency, sustainability, and cost reduction [9].

The research describes how the HEMS can assist the prosumers in controlling both energy production and consumption particularly as there will be more dependence on renewable sources. Methods are categorised into four broad groups: traditional optimization methods, model predictive control, heuristics and metaheuristics, and others based on AI, and some of the main computational and technical issues, which include uncertainty, user behaviour, planning horizons, and the complexity of models, are discussed. The authors demonstrate that the latest developments such as metaheuristic algorithms, reinforcement learning, and forecasting models have enhanced cost saving, energy efficiency, and system stability in residential energy management. The general view of the work is an amalgamation of advances in methods and issues to provide a comprehensive insight into advancement in the research of HEMS [10].

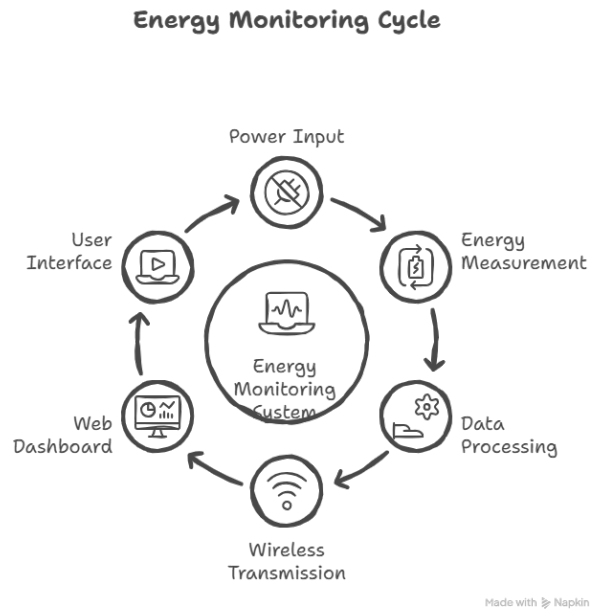

Fig. 1. System Architecture

III. SYSTEM DESIGN

A. Problem Definition

The solution to effective energy monitoring has become very popular in residential buildings especially in the past few years due to the high rate of growth in electricity consumption. Traditional energy meters usually only give cumulative readings, and they do not offer real-time measurements of the electrical parameters, e.g. the voltage, current, power and energy consumption. Users usually do not know the real-time electricity consumption pattern and cost of it immediately because it cannot be viewed instantaneously. This weakness complicates the ability to notice abnormal consumption, control energy and inaccurate estimation of electricity costs. Moreover, most of the currently available smart monitoring systems are costly, or they demand sophisticated infrastructure, which might not be applicable to small residential applications. Thus, it is necessary to have an affordable, Wi-Fi-based smart energy monitoring solution, capable of delivering real-time measurements and visualization, as well as simple cost estimation to a basic dashboard interface. This work aims to create and deploy such a system with the help of ESP32 microcontroller and PZEM-004T energy sensor to enhance the level of energy awareness and monitoring performance in residential conditions.

B. System Architecture

(fig 1) The proposed smart energy monitoring implementation system is developed based on an integrated hardware and software platform to obtain real-time information and visualize it. The system will be composed of an ESP32 microcontroller and PZEM-004T energy measuring module. The sensor allows measuring the main electrical parameters, such as voltage, current, power, and total energy consumption, which is PZEM-004T. These measurements are sent to the ESP32 by the serial communication. The central processing unit of the system is the ESP32, which reads the sensor data, processes it and makes it ready to be transmitted. With an in-built Wi-Fi, the ESP32 data is connected to a local wireless network and transfers the information on the readings to a web-based dashboard interface.

$$P = V \times I \tag{1}$$

$$E = P \times t \tag{2}$$

Real-time visualization of electrical parameters is offered by the dashboard, which enables the user to observe the real values of live voltage, current, power(1), and energy consumption(2). Besides monitoring, the system has a balance estimation service which determines electricity cost depending on the amount of total energy used and a value which is set as unit price. The approximated price and balance is indicated in the dashboard so that the user can know how much he spends on electricity. Any 5 V power supply is regulated to provide a stable and reliable operation of the systems. The general design of the system is quite simple, extendable, and inexpensive, and thus it can be used in smart home energy monitoring applications.

C. Required Hardware Components

ESP32 Microcontroller(Fig. 2) Function: Plays the role of processing unit of the system. Role: Gathers information on the PZEM-004T sensor, processes information, and sends the measurements to the server through Wi-Fi.

Fig. 2. Architecture of ESP MicroController

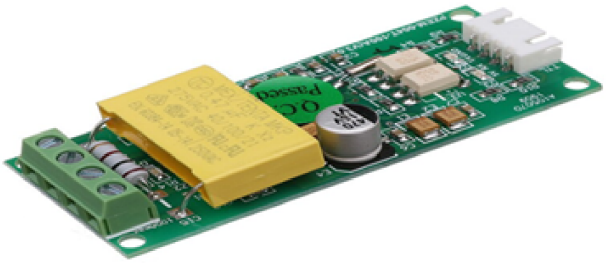

Fig. 3. PZEM-004T Sensor Architecture

PZEM-004T(Fig. 3)Energy Meter Module Function: Elevation of electrical parameters - current, energy consumption, voltage, and power.
Relay Module (Dual Channel) Function: Automatically or manually switches electrical loads (e.g. bulb, fan) based on AI forecasts.
Load Devices (Test Appliances) Examples: Bulb, or any low/medium power appliance. Function: Acts as a test load for predictive control. Power Supply Specification: 5V Function: ESP32, relay, and sensor module powered.

Cables and Connectors Includes: Jumper wires, USB cable for programming, screw terminals for PZEM connection. Function: Ensures reliable communication between all components.

Breadboard or PCB Function: This is used to prototype or to assemble the final circuit
Wi-Fi Router (or Mobile Hotspot) Function: Available to provide network connectivity to transmit data to the backend. Display Module Function: Displays local real-time numbers (voltage, current, power).

D. System Implementation

(fig 4) To achieve this, the proposed smart energy monitoring system was executed through hardware and software modules that were put together to give real-time monitoring and visualization of electrical parameters. The hardware used will include ESP32 microcontroller and PZEM-004T energy measurement module. The PZEM-004T sensor was attached to the main electrical line with the aim of measuring the voltage, current, power and cumulative energy used. Serial communication (RX/TX) is used to communicate with the sensor by ESP32. Controlled supply of 5V was applied to get a stable work with ESP32 and other components. The ESP32 was programmed in Arduino IDE platform. Serial communication was also set up in the firmware to constantly retrieve the values of electrical parameters of the PZEM-004T module. The ESP32 was used to process the raw sensor data to provide a voltage (V), current (A), power (W), and energy (kWh) values. The cost of the electricity was then determined by the total amount of energy used and a unit price that was set by the microcontroller. The calculation of the balance was done by deducting the computed cost to an initial balance value and all where displayed in LED display(fig 5) .
To communicate, the Wi-Fi embedded module of the ESP32 was used in order to connect the device to a local wireless network. The dashboard was created in the form of a web-based tool that is used to show the real-time data of electrical parameters and cost-related data. The ESP32 sent or hosted data to the dashboard interface and allowed users to read live data using a computer or mobile device using a web browser.(fig 7) The dashboard was made in such a way that it shows voltage, current, power, total energy used, estimated cost, and the remaining balance in a well-arranged and easy to use format.
The system was made to be linked with household electrical loads like bulbs and fans during the testing process to monitor real time changes in power consumption. The dashboard was able to display the variations in voltage, current and power with regard to changes in load. The cost and balance value was dynamically calculated depending on the consumption of energy. The application was shown to be stable in performance, measuring, and working steadily with wireless communication.

IV. RESULTS AND DISCUSSION

A. Results

The Smart Energy Monitoring System proposed using Wi-Fi was implemented and tested successfully in case of real-time residential loading conditions. The hardware component comprised of ESP32 microcontroller, PZEM-004T energy sensor, TFT/LCD display and relay module worked well during the experiment. The system was linked with some domestic loads like a bulb and small domestic appliances to monitor the changes in electrical parameters. The sensor, PZEM-004T was able to measure voltage, current, power and cumulative energy consumption accurately. Under normal operations conditions testing voltage was noted to be around 228 230 V. The values

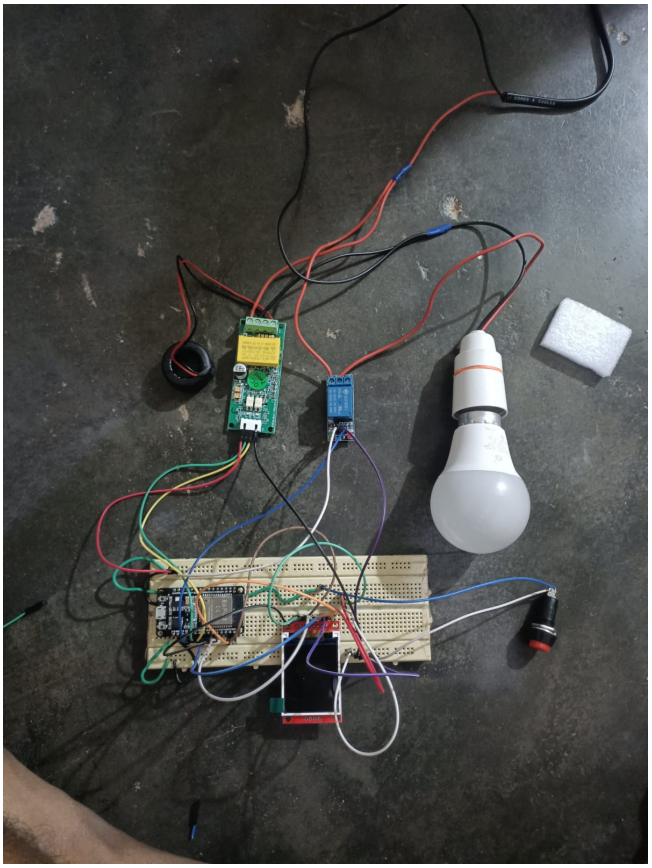

Fig. 4. Implementation of the System

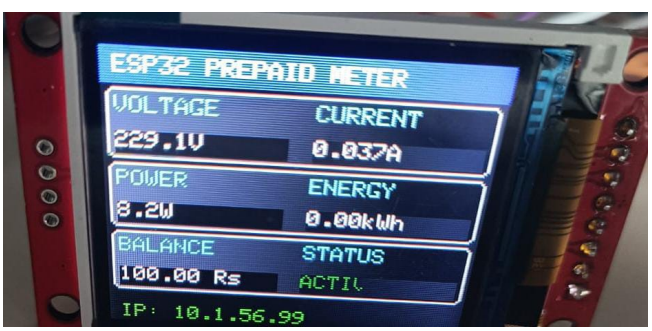

Fig. 5. Results shown in display

of current and power changed proportionally to the variation in connected loads. Indicatively, the power consumption was measured at about 8-10 W when a low power load was attached and it then grew slowly in kWh units. Such changes were managed to be displayed in real-time on the local TFT screen(fig 6) and on the web-based dashboard. The local display module offered real-time visualization of parameters of the system such as voltage, current, power, energy, amount of balance, and IP address of the device. There was also an alert interface that indicated predetermined threshold conditions like over-voltage limit, over-current limit, theft current detection and minimum balance level. The warning display screen

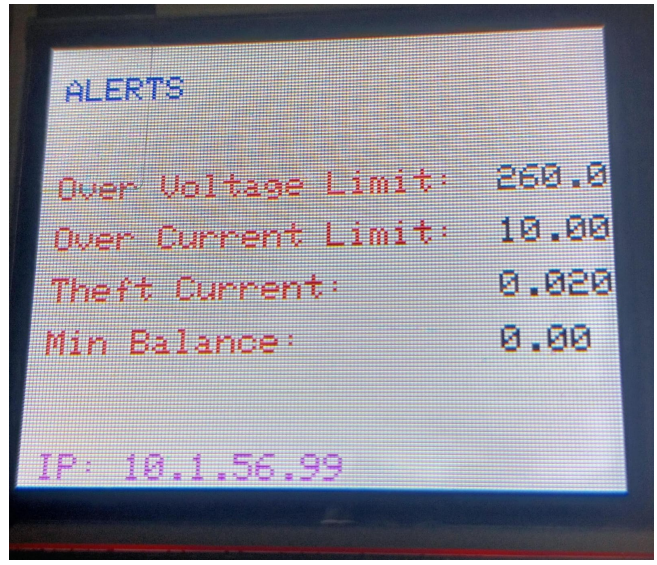

Fig. 6. Alerts Displayed if theft found

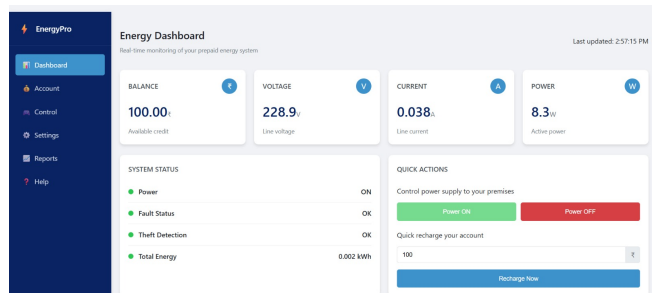

Fig. 7. Dasboard Representation

proved that the system is expandable to safety observance and the detection of abnormal conditions.

The web dashboard interface monitored real time electric parameters and system status indicators. Balance, voltage, current, power, total amount of consumed energy, and system status [Power ON/OFF, Fault Status, Theft Detection] were displayed on the dashboard. The Power ON and Power OFF control buttons were also tested successfully and they displayed the ability to control remotely through the Wi-Fi. The dashboard was available on both desktop and mobile browsers which affirmed that it was cross-platform. The balance estimation(4) feature worked as anticipated. The system estimated the cost of energy(3) as:

$$Cost = E \times Tariff(Costperunitofelectricity) \quad (3)$$

$$Balance_{new} = Balance_{old} - Cost \quad (4)$$

The rest balance was updated dynamically depending on the consumption of energy. In the case of testing, as the energy increased, the balance value correspondingly reduced in proportion. It proves that the logic of calculating costs and displaying it was properly implemented. ESP32 to the web dashboard communication over Wi-Fi was also stable

during testing. The data refresh rate was also high enough to monitor real time without any apparent lag. There are no significant communication breakdowns noticed in the local network environment. All in all, the experiments show that the proposed system is accurate in measurement, transmits wirelessly, has real-time visualization, and calculates balance. The system is able to meet the goal of offering a cost-effective and scalable smart energy monitoring system to the residential setting. Also, the modular architecture provides expansion opportunities in the future like automatic cutoff, cloud integration, advanced analytics, and mobile application support.

B. Conclusion

The design and implementation of the Wi-Fi based Smart Energy Monitoring System, which is based on an ESP32 microcontroller and PZEM-004T energy sensor, were presented in this paper. The system has been able to gauge and monitor critical electrical values such as voltage, current, power and total energy consumed in real time. The obtained measurements are wirelessly sent by the in-built Wi-Fi chip on the ESP32 and shown on a conveniently designed web dashboard interface.

The designed system allows the user to see the live electrical parameters, overall energy consumption, and approximate the cost of electricity depending on the consumption. The balance estimation feature offers greater awareness on the energy usage and promotes the use of energy efficiently. Also, there is the introduction of local LCD/TFT display and alert functionality, which makes the use of the system easier and more reliable.

The experimental findings prove that the suggested system can be successful and provide stable communication in the environment of a local network. The dashboard displays the dynamic results of the load conditions, the cost calculation system is efficient. The general design is cheap, scalable and appropriate in home-based smart energy monitoring systems.

C. Future Work

Even though the proposed Wi-Fi based Smart Energy Monitoring System manages to deliver real-time monitoring and estimate of the balance, some improvements can be made in future in order to enhance the functionality, scalability, and intelligence. A significant enhancement would include incorporating the cloud-based data storage. At present, the system is used in a local Wi-Fi environment. With the incorporation of cloud platforms, it would become possible to store data in the long-term and access it remotely but from any place. This would enable users to study the past consumption and provide detailed reports.

The introduction of automatic power stop in case the balance hits zero or in case of abnormal conditions including over-voltage or over-current could be another improvement. Although alert mechanisms have been installed, it would be better to automate protective measures to enhance the system safety and make them function like a commercial prepaid smart meter.

They can also be advanced with machine learning and data analytics that would forecast the future consumption of energy, based on past usage behaviors. Predictive analysis would assist the user to control energy better and save on electricity bills.

Specific mobile app can be created to enhance accessibility among the users and real-time notification to alert them about low balance, abnormal current flow, or fault conditions. User awareness and responsiveness would be improved with the help of push notifications. Moreover, it is possible to expand the system to multi-load monitoring, three-phase power systems, and connection with renewable sources of energy like solar panels. This would ensure that the system was made fit with industrial and big scale uses.

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